q2_shortcircuit.c - Code::Blocks / VS Code (C)

```c
#include <stdio.h>

static long calls = 0;
int slow_verify(int id) { calls++; return id % 10 == 0; }
int is_senior(int id)   { return id % 4 == 0; }
int class_ok(int id)    { return id % 2 == 0; }

int main(void)
{
    int n = 1000, hits;

    calls = 0;
    int r1 = 0 && slow_verify(1);
    int r2 = 1 || slow_verify(1);
    printf("Demo: calls after (0 && f) and (1 || f) = %ld (r1=%d r2=%d)\n", calls, r1, r2);

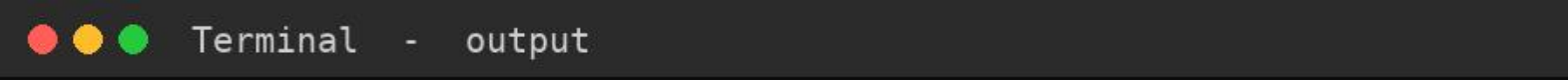
    calls = 0; hits = 0;
    for (int id = 1; id <= n; id++)
        if (slow_verify(id) || (is_senior(id) && class_ok(id))) hits++;
    printf("A slow first          : calls=%ld concessions=%d\n", calls, hits);

    calls = 0; hits = 0;
    for (int id = 1; id <= n; id++)
        if ((is_senior(id) && class_ok(id)) || slow_verify(id)) hits++;
    printf("B cheap first         : calls=%ld concessions=%d\n", calls, hits);

    calls = 0; hits = 0;
    for (int id = 1; id <= n; id++)
        if ((is_senior(id) & class_ok(id)) | slow_verify(id)) hits++;
    printf("C bitwise & and |     : calls=%ld concessions=%d\n", calls, hits);

    calls = 0; hits = 0;
    for (int id = 1; id <= n; id++) {
        if (is_senior(id)) {
            if (class_ok(id)) { hits++; continue; }
        }
        if (slow_verify(id)) hits++;
    }
    printf("D nested if           : calls=%ld concessions=%d\n", calls, hits);

    int x = 0, y = 0;
    if (0 && (++x)) {}
    if (0 &  (++y)) {}
    printf("&& skipped increment: x=%d ; & did not skip: y=%d\n", x, y);
    return 0;
}
```


Terminal - output
user@pc:~/Assignment2$ gcc -Wall q2_shortcircuit.c -o q2 && ./q2
Demo: calls after (0 && f) and (1 || f) = 0 (r1=0 r2=1)
A slow first : calls=1000 concessions=300
B cheap first : calls=750 concessions=300
C bitwise & and | : calls=1000 concessions=300
D nested if : calls=750 concessions=300
&& skipped increment: x=0 ; & did not skip: y=1
user@pc:~/Assignment2$ _